

Automated Twice-Daily Content Engine

From live topic discovery to published YouTube Short in a fully unattended pipeline

09:00 IST

21:00 IST

Free Tooling

No Manual Ops

What It Does

- Selects topic (rotate or trending)
- Fetches latest relevant features
- Generates structured script with Gemini
- Synthesizes voice and captions
- Renders animated vertical video
- Uploads to YouTube with metadata and thumbnail

Architecture Flow

```
GitHub Actions (15-min poll)
→ Gate (IST slot + de-dup)
→ Test (pytest)
→ Publish Stages:
  1) Script  2) Voice
  3) Render  4) Upload
→ history.json commit
```

External integrations:

- Gemini (google-genai)
- edge-tts
- Remotion + ffmpeg
- YouTube Data API v3

Business value: predictable publishing cadence, low operational cost, auditable automation.

Detailed Pipeline and Control Points

- 1 Topic Strategy** Rotate or trending mode with cooldown.
- 2 Feature Fetch** RSS + Google News, then freshness and relevance filtering.
- 3 Script AI** Gemini generates strict JSON: title, lines, points, flow.
- 4 Voice + Timing** edge-tts creates voice.mp3 and captions.
- 5 Animation Render** Remotion renders vertical out.mp4 from typed props.
- 6 YouTube Publish** Resumable upload with retry and metadata.

Governance & Reliability

- Gate windows: 09:00-12:00 and 21:00-00:00 IST
- Exactly one successful run per slot (de-dup)
- Stage separation improves observability
- Typed exceptions + retries for transient failures
- No database, auditable state in history.json

Audience Outcomes

- **Senior Managers:** predictable output and cost control
- **Enterprise Architects:** clear contracts and resilient orchestration
- **Development Managers:** isolated stages and faster troubleshooting

End result: **twice-daily, fully automated Shorts publishing.**